12th BOARD HACKER JUGAAD NOTES

An Optimization Problem :-

An optimization problem is a problem which seeks to maximize or minimize a function is called an optimization problem. An optimization problem may involve maximization of profit, production etc. or minimization of cost, from available resources etc.

A linear programming problem (LPP) :-

A linear programming problem deals with the optimization (maximization/minimization) of a linear function of two variables (say x and y) known as objective function subject to the conditions that the variables are non-negative and satisfy a set of linear inequalities (called linear constraints). A linear programming problem is a special type of optimization problem

Objective Function :-

Linear function $Z = ax + by$ where a and b are constants, which has to be maximized or minimized is called a linear objective function

Decision Variables :-

In the objective function $Z = ax + by$, x and y are called Decision variables.

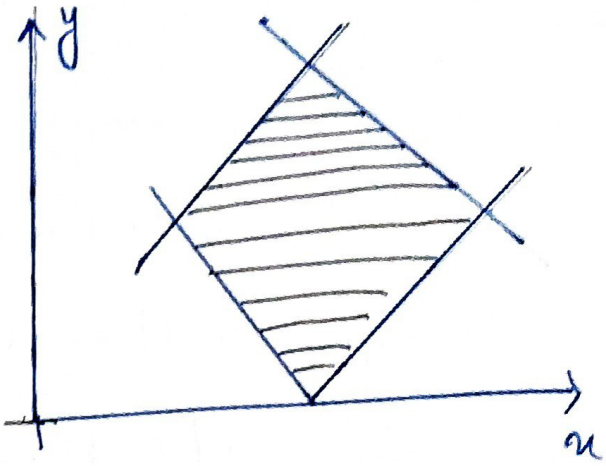
Constraints :-

The linear inequalities or restriction on the variables of an LPP are called Constraints. The condition $x \geq 0, y \geq 0$ are called non-negative constraints. non-negative constraints $x \geq 0, y \geq 0$ of an LPP is called the feasible region for the problem.

$$3x + 5y \geq 7$$

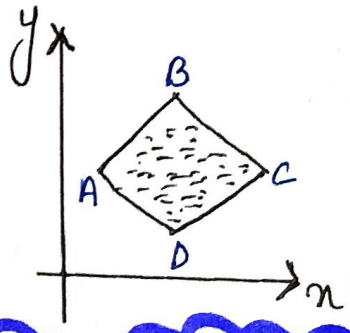
Feasible Region :-

The common region determined by all the constraints including non-negative constraints $x \geq 0, y \geq 0$ of an LPP is called the feasible region for the problem.



Feasible solutions :-

Points within and on the boundary of the feasible region for an LPP represent feasible solutions.



Infeasible Solutions :-

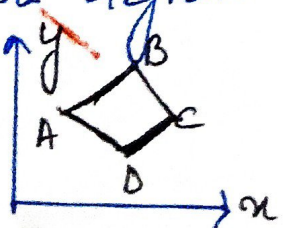
Any point outside feasible region is called an Infeasible solution.

Optimal (feasible) Solution:-

Any point in the feasible region that gives the optimal value (maximum or minimum) of an objective function is called an optimal solution

Theorem 1:-

Let R be the feasible region (convex polygon) for an LPP and let $Z = ax + by$ be the objective function. When Z has an optimal value (maximum or minimum) where x, and y are subject to constraints described by linear inequalities, optimal value must occur at a corner point (vertex) of the feasible region.



Theorem - 2

Let R be the feasible region for a LPP and let $Z = ax + by$ be the objective function. If R is bounded, then the objective function Z has both a maximum and a minimum value on R and each of these occur at a corner point of R.

If the feasible region R is unbounded, then the maximum or a minimum value of the objective function may or may not exist. However, if it exists, it must, it must occur at a corner point of R.

Corner point method for Solving a LPP:-

The method comprises of the following steps:-

1) Find the feasible region of the LPP and determine its corner points (vertices) either by inspection or by solving the two equations of the the lines intersecting at that point.

2) Evaluate the objective function $Z = ax + by$ at each corner point.
Let M and m, respectively denote the largest and the smallest value of Z.

3) (i) When the feasible region is bounded, M and m, are, respectively, the maximum and minimum values of Z.

(ii) In case, the feasible region is unbounded

(a) M is maximum value of Z, if the open half plane determined by $ax + by > M$ has no point in common with the feasible region. otherwise, Z has no maximum value.

b) Similarly, m is the minimum of Z, if the open half plane determined by $ax + by < m$ has no point in common with the feasible region. Otherwise, Z has no minimum value.

Multiple optimal points :-

If two corner point of the feasible region are optimal solutions of the same type. i.e both produce the same maximum or minimum, then any point on the line segment joining these two points is also an optimal solution of the same type.

